Project Development Phase
Data Visual

Date	06 March 2026
Team ID	
Project Name	**Strategic Product Placement Analysis:** Unveiling Sales Impact with Tableau Visualization
Maximum Marks	3 Marks

PREPROCESSING STEPS

1. Data Import and Initial Inspection

The dataset related to product positioning and sales performance was imported into Tableau Public for analysis and visualization.

The dataset consisted of 1000 records and 10 attributes:

- Product ID
- Product Position
- Price
- Competitor's Price
- Promotion
- Foot Traffic
- Consumer Demographics
- Product Category
- Seasonal
- Sales Volume

An initial inspection was performed to understand the structure, data types, and completeness of the dataset. This helped identify the variables that influence sales performance and product placement strategies.

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2. Handling Missing Values

The dataset was examined to identify **null or missing values** across all attributes.

Observation:
No significant missing values were detected in the dataset.

Since the dataset was complete, **no imputation or deletion was required** for missing values. This ensured accurate analysis and reliable visualization results.

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3. Removal of Duplicate Records

The dataset was checked for duplicate records to ensure data integrity.

Duplicate entries were removed to ensure:

- Accurate sales aggregation
- Reliable analysis of product performance
- Prevention of inflated sales values

Removing duplicates improved the reliability and accuracy of the analytical results.

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4. Data Classification (Dimension & Measure Separation)

Fields were categorized appropriately in **Tableau**.

Dimensions

- Product ID
- Product Position
- Promotion
- Consumer Demographics
- Product Category
- Seasonal

Measures

- Price
- Competitor's Price
- Foot Traffic
- Sales Volume

This classification enabled proper aggregation and effective visualization of the dataset.

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5. Product Position Analysis

Product positioning was analyzed to understand how placement strategies affect sales performance.

Products were categorized based on their placement positions such as:

- Front or premium positions
- Regular shelf positions
- Promotional placements

This helped determine which placement strategies attract more customer attention and increase sales volume.

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6. Pricing and Competitor Comparison

The dataset includes product price and competitor price, allowing comparison between pricing strategies.

This analysis helped identify:

- Price competitiveness in the market
- Relationship between product pricing and sales volume
- Impact of competitor pricing on sales performance.

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7. Promotion Impact Analysis

The Promotion attribute indicates whether a product was part of a marketing campaign.

This analysis helps evaluate:

- The effectiveness of promotional strategies
- Whether promotions significantly increase sales
- Which products benefit most from promotional activities.

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8. Consumer Behavior Analysis

The Consumer Demographics field helps analyze customer purchasing behavior.

This allowed segmentation of consumers based on:

- Demographic groups
- Purchasing patterns
- Response to promotions and product placement.

Understanding consumer behavior helps businesses design more effective marketing and placement strategies.

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9. Seasonal Sales Analysis

The Seasonal attribute helps analyze how different seasons influence product demand.

This allows businesses to identify:

- Seasonal purchasing patterns
- High-demand periods
- Seasonal marketing opportunities.

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10. Foot Traffic Analysis

Foot traffic data helps measure the number of customers visiting a store or platform.

This analysis helps determine:

- The relationship between customer traffic and sales volume
- Whether higher traffic leads to higher product sales
- Which product placements attract more customers.

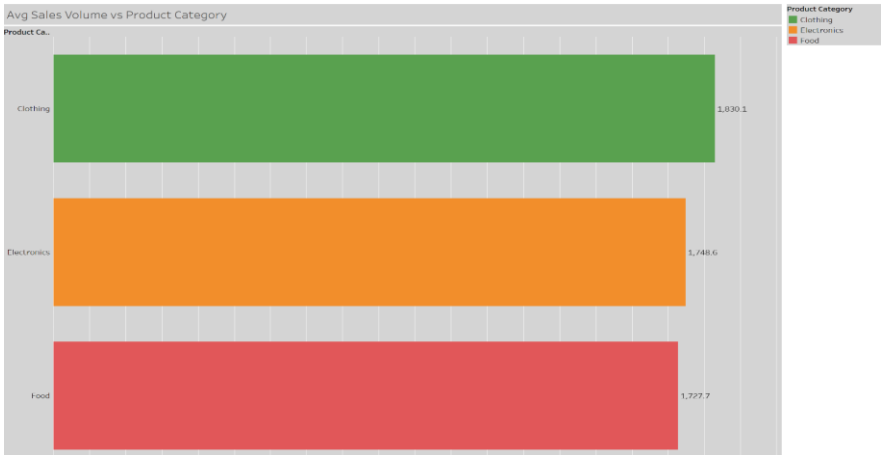
BUSINESS QUESTIONS ADDRESSED

The following business questions guided the analysis of **Strategic Product Placement and Sales Performance**.

1. Average Sales Volume by Product Category

Business Questions:

- Which product categories generate the highest sales volume?
- Are certain product categories more popular among customers?



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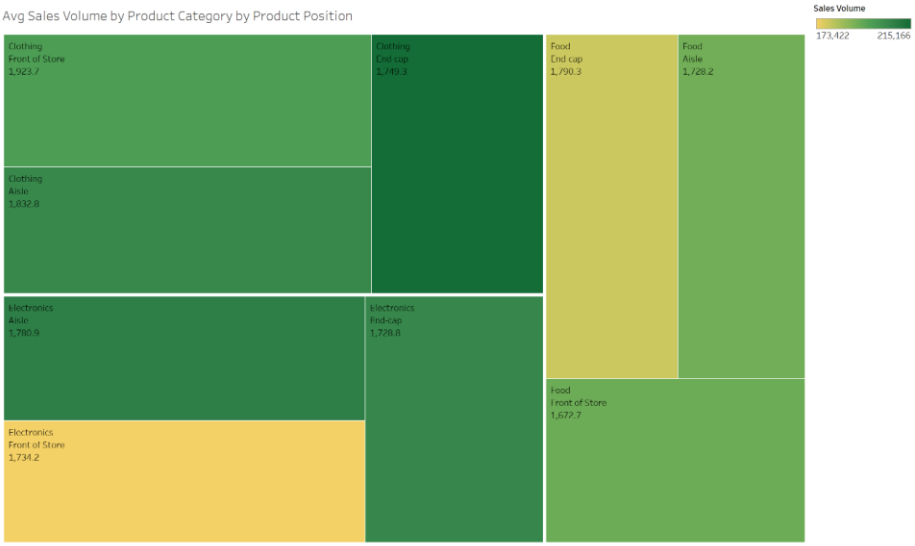
2. Sales Performance by Product Position

Business Questions:

- How does product placement affect sales volume?
- Which product positions generate the highest sales?



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3. Foot Traffic vs Sales Volume

Business Questions:

- Does higher customer traffic lead to increased sales?
- How strongly is sales volume related to foot traffic?

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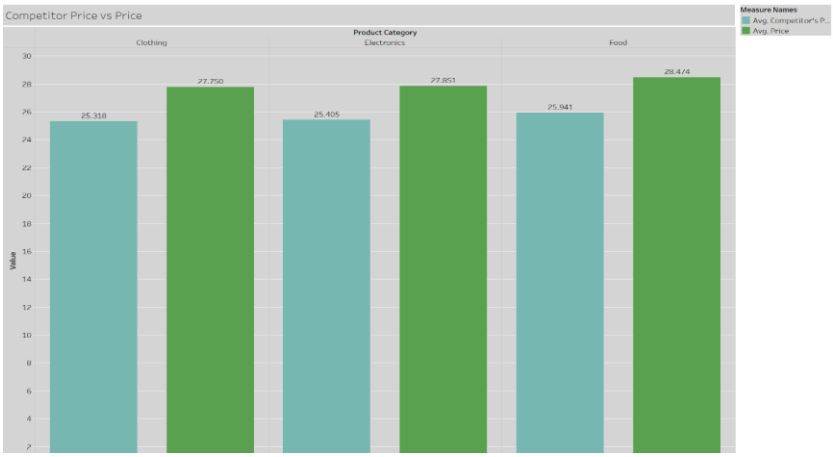
4. Price vs Competitor Price Comparison

Business Questions:

- How does product pricing compare with competitor pricing?
- Does pricing strategy influence product sales?

- 

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5. Promotion Impact on Sales

Business Questions:

- Do promotional campaigns increase sales performance?
- Which products benefit the most from promotions?

Promotion of Product Category on Price and Sales Volume

Promotion	Product Catego.	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

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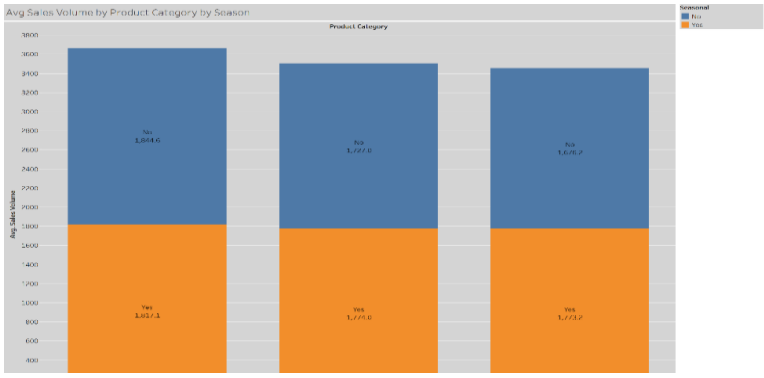
6. Seasonal Impact on Product Sales

Business Questions:

- Do seasonal factors influence customer purchasing behavior?
- Which seasons generate the highest sales?



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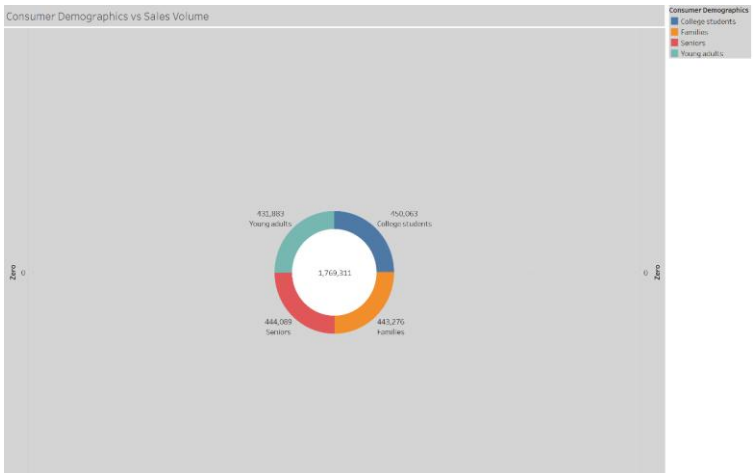
7. Consumer Demographics and Purchasing Patterns

Business Questions:

- How do different consumer groups respond to product placements?
- Which demographic segments contribute the most to sales?
- 

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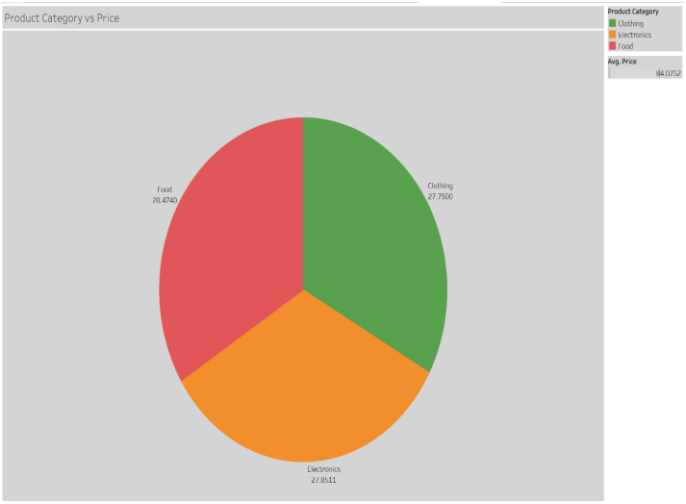
8. Product Category vs Price

Business Questions:

What is the price distribution across different product categories?

Which product categories have the highest and lowest average prices?


Product Category vs Price
Product Category
Clothing
Electronics
Food
Avg. Price
84.0752
Food
28.4740
Clothing
27.7500
Electronics
27.8511